AWS VPC PEERING

FINAL PROJECT REPORT

EWB Courses — AWS Cloud Computing Internship

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Project	VPC Peering Between Two VPCs
AWS Region	Asia Pacific (Mumbai) — ap-south-1
VPC A	vpc_A — 10.10.0.0/16
VPC B	vpc-b — 10.20.0.0/16
Peering connection	ewb-a-b-peer — pcx-05a080f5599625be7
Project evidence date	27 October 2026

This report organizes the AWS console and terminal screenshots captured during the project, with an overview of the VPC design, routing, security groups, EC2 inventory and connectivity test.

1. Project Objective

The purpose of this project is to configure VPC peering between two independent Amazon VPCs and verify private network communication between application instances. The two VPCs use non-overlapping IPv4 CIDR blocks. Route tables direct remote-VPC traffic through the peering connection, while security groups control SSH and ICMP access.

2. Architecture and Resource Inventory

Component	Configuration / observed details
Region	ap-south-1 — Mumbai
VPC A	vpc_A — 10.10.0.0/16
VPC B	vpc-b — 10.20.0.0/16
App-A	ewb-app-a — private IP 10.10.2.10
App-B	ewb-app-b — private IP 10.20.1.10
Bastion	ewb-bastion — private IP 10.10.1.10; public IPv4 43.205.237.20
Peering	ewb-a-b-peer — pcx-05a080f5599625be7 — Active

3. Network and Route Design

VPC A uses 10.10.0.0/16 and VPC B uses 10.20.0.0/16. The private application subnets are 10.10.2.0/24 (VPC A) and 10.20.1.0/24 (VPC B). The VPC A private route table contains a route for 10.20.0.0/16 through the peering connection. The VPC B private route table contains a route for 10.10.0.0/16 through the same peering connection. Both peering routes are shown as Active in the supplied console screenshots.

4. Security Controls

The App-A security group screenshot shows SSH TCP/22 and ICMP Echo Request inbound rules sourced from 10.10.1.10/32. The App-B security group screenshot shows ICMP Echo Request sourced from 10.10.2.10/32 and SSH TCP/22 sourced from 103.76.189.250/32. These source restrictions are documented as displayed in the console screenshots.

5. Connectivity Verification

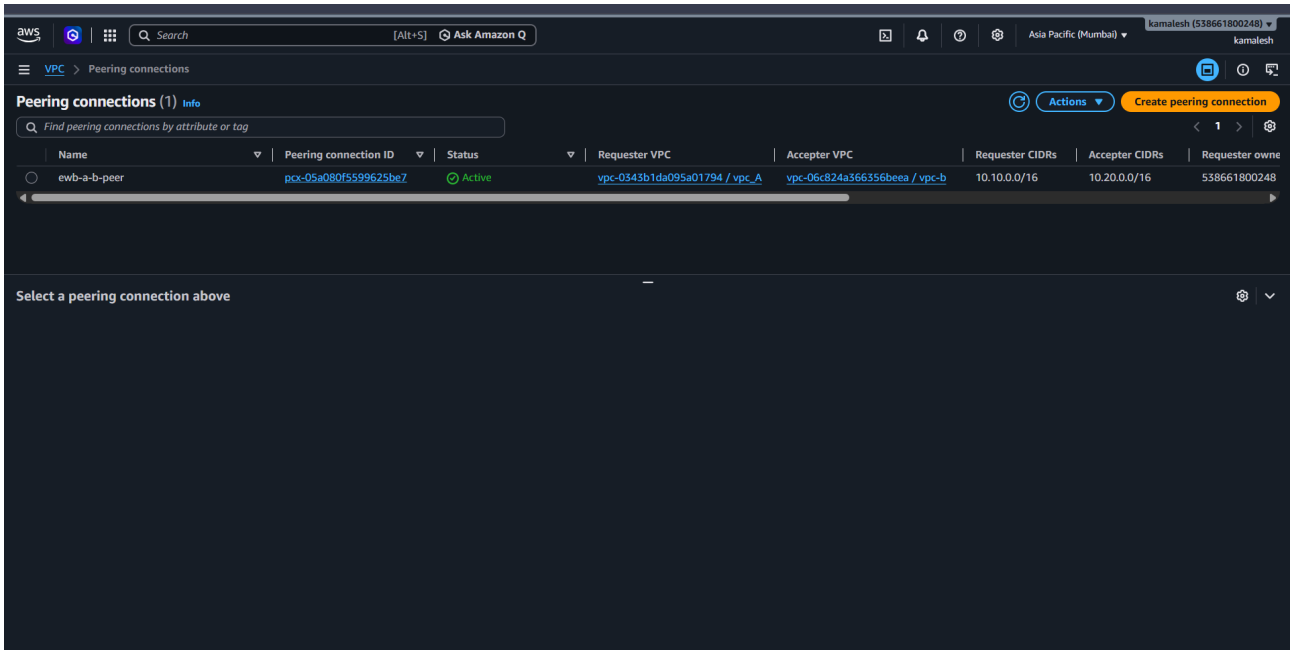
The supplied terminal screenshot records a ping from App-A (10.10.2.10) to App-B (10.20.1.10). The successful run shows 4 packets transmitted, 4 received and 0% packet loss. This is evidence that ICMP connectivity was working in that direction at the time of the test.

6. Conclusion

The available evidence documents the two VPC CIDR ranges, an Active VPC peering connection, the private route-table entries, subnet associations, EC2 instance status and security-group rules. The successful App-A-to-App-B ping demonstrates private connectivity through the peering route. The screenshots below are arranged as supporting project evidence.

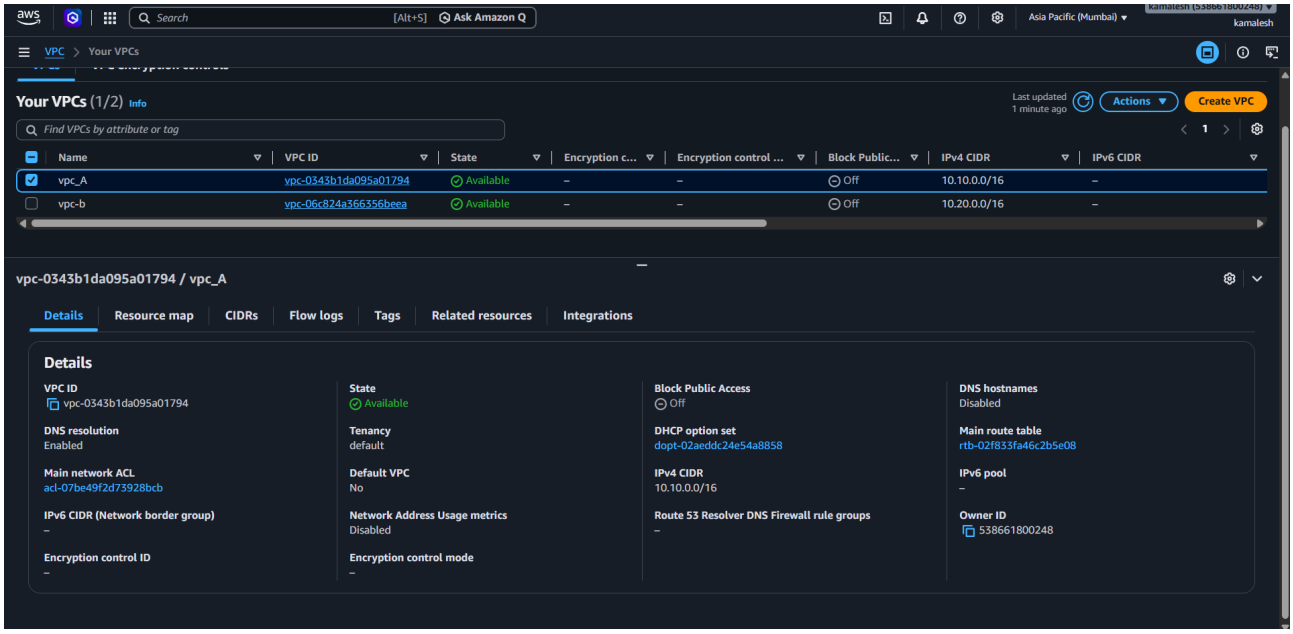
Evidence 1 — VPC Peering Connection

Shows ewb-a-b-peer in Active state and the requester/acceptor VPC CIDR ranges.



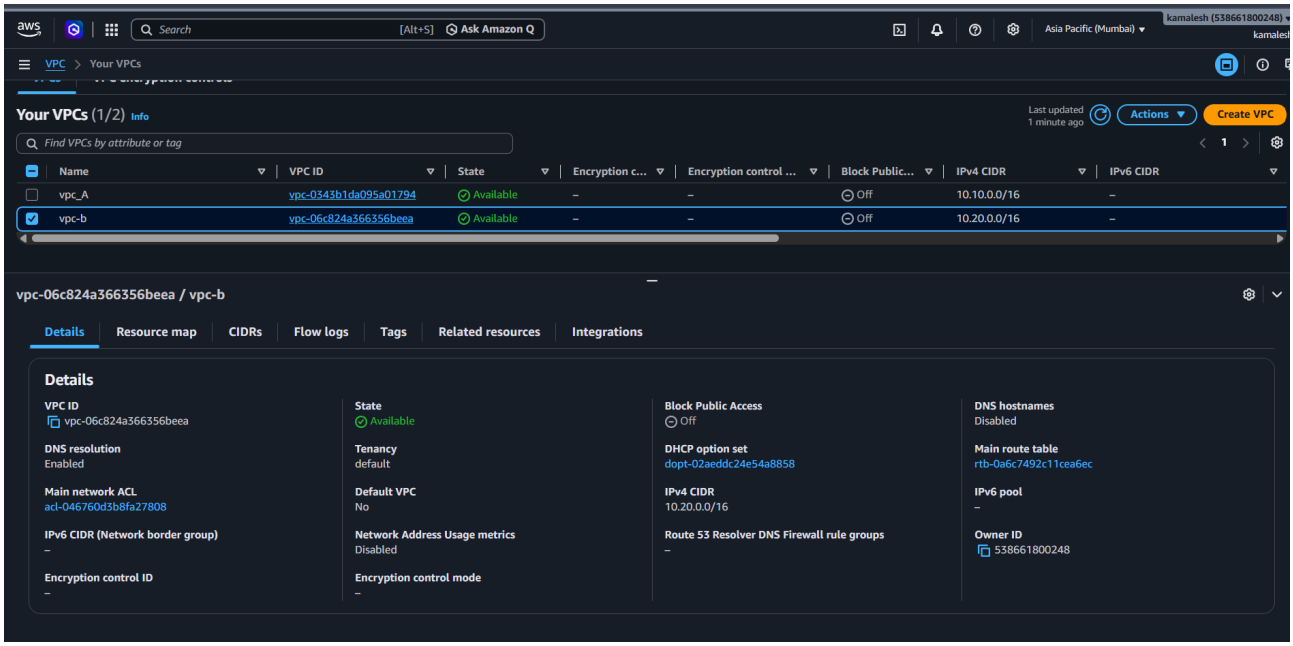
Evidence 2 — VPC A Details

Shows vpc_A available with CIDR 10.10.0.0/16.



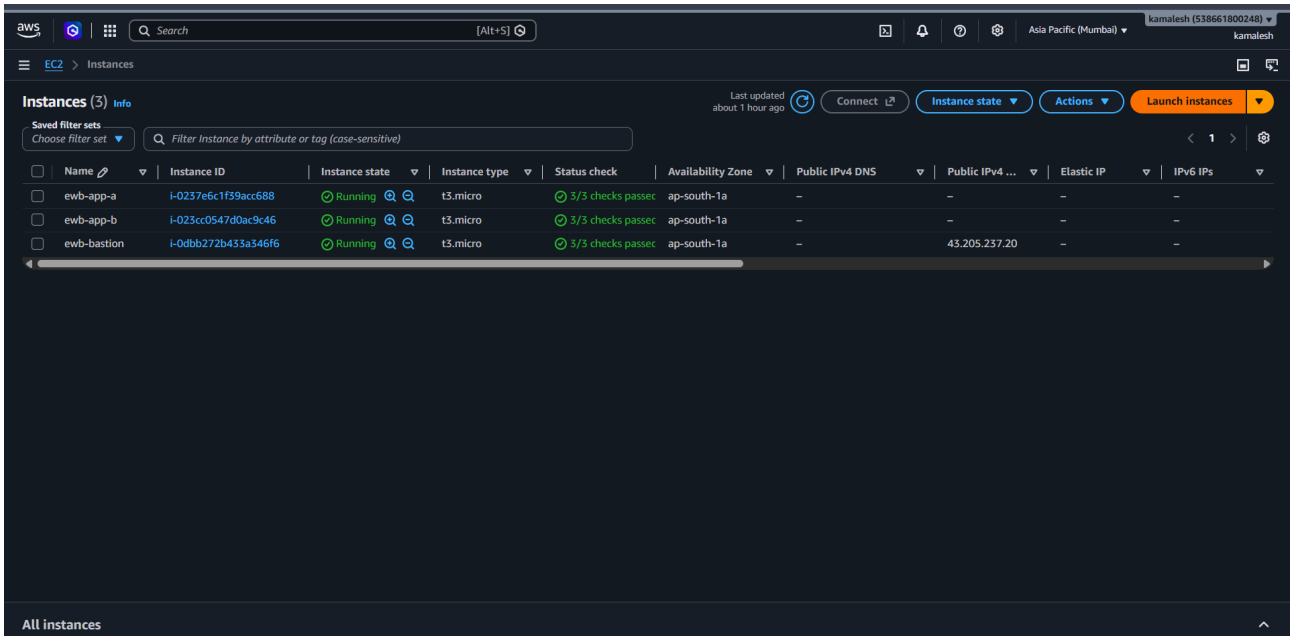
Evidence 3 — VPC B Details

Shows vpc-b available with CIDR 10.20.0.0/16.



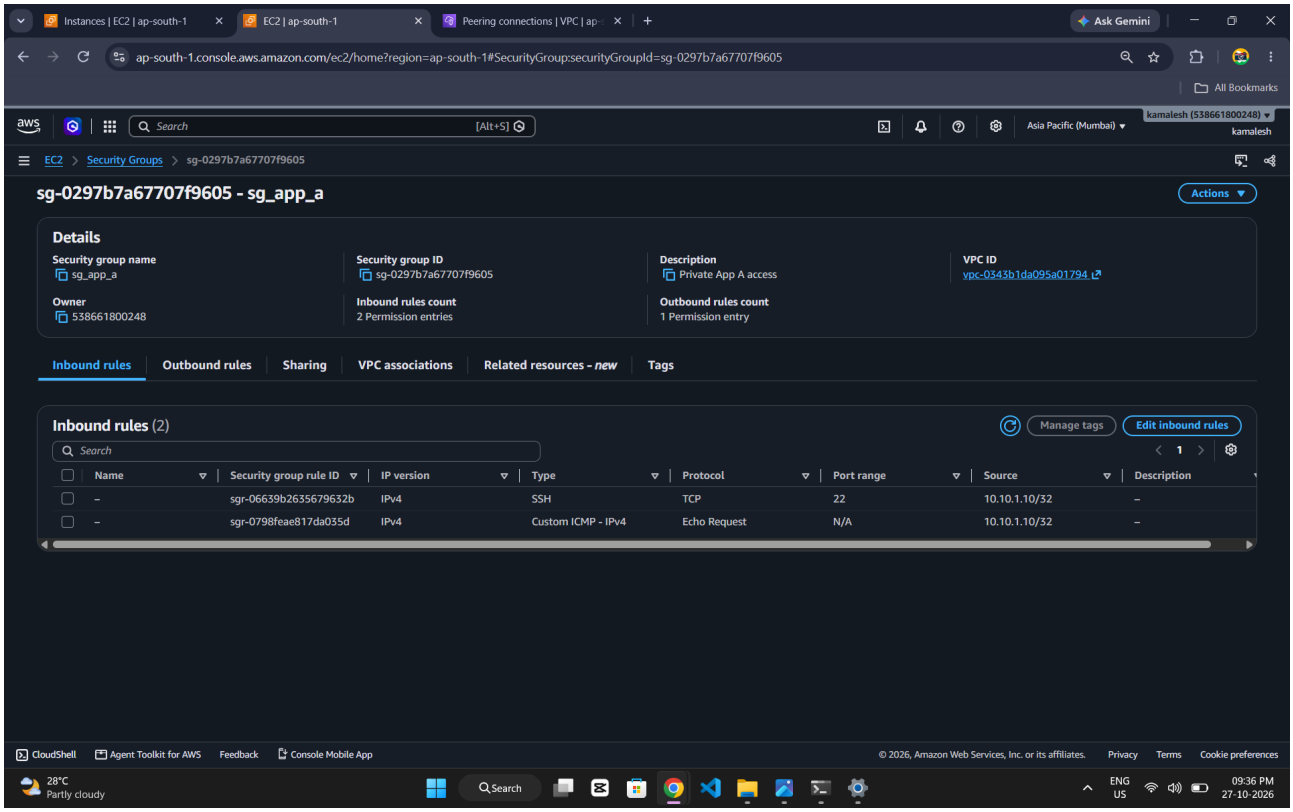
Evidence 4 — EC2 Instance Inventory

Shows ewb-app-a, ewb-app-b and ewb-bastion running; App-A and App-B have no public IPv4 address displayed.



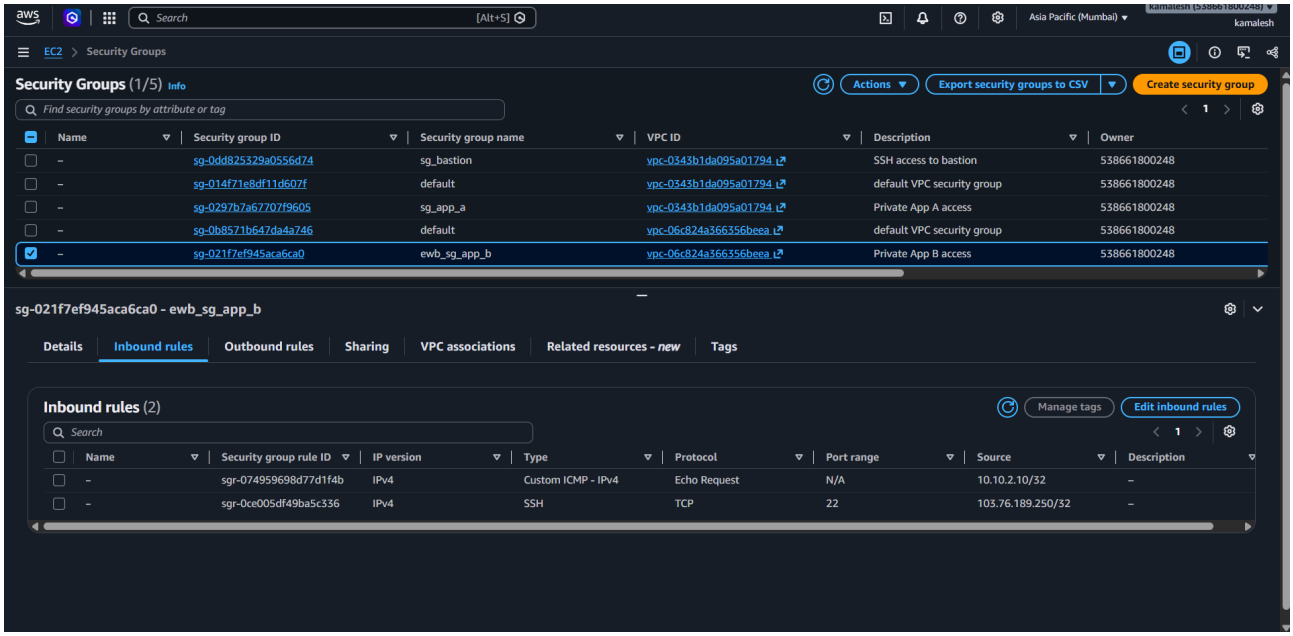
Evidence 5 — App-A Security Group

Shows App-A inbound SSH and ICMP Echo Request rules, sourced from 10.10.1.10/32.



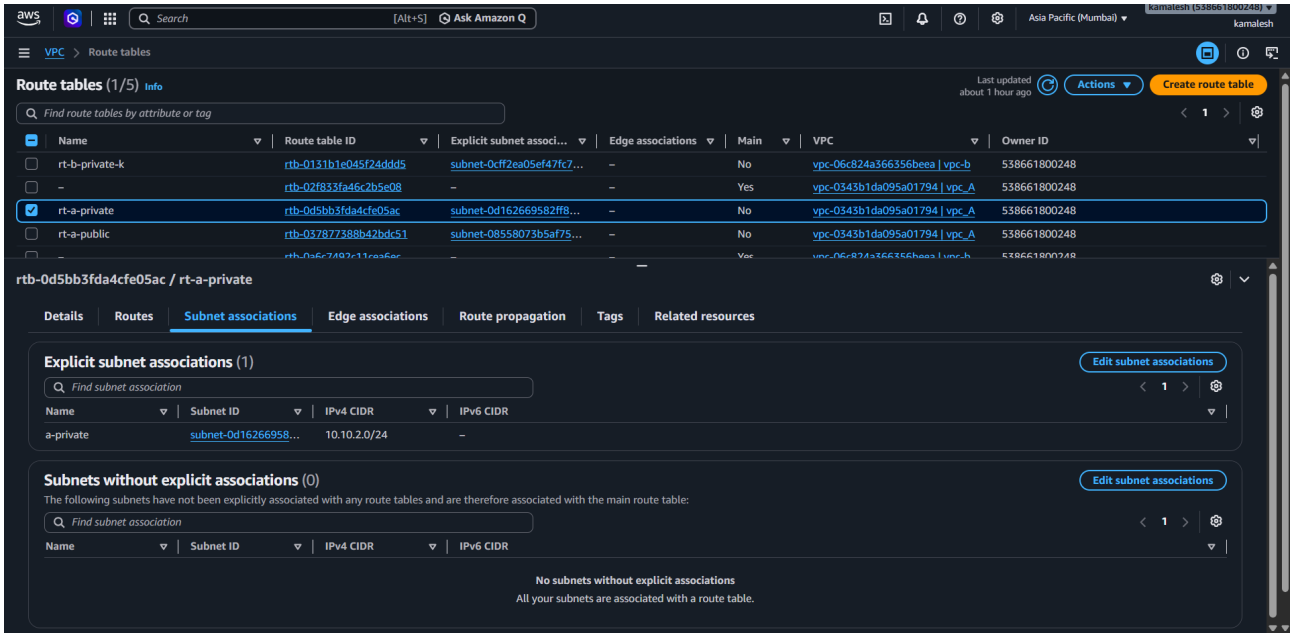
Evidence 6 — App-B Security Group

Shows App-B inbound ICMP Echo Request from 10.10.2.10/32 and SSH TCP/22 from 103.76.189.250/32.



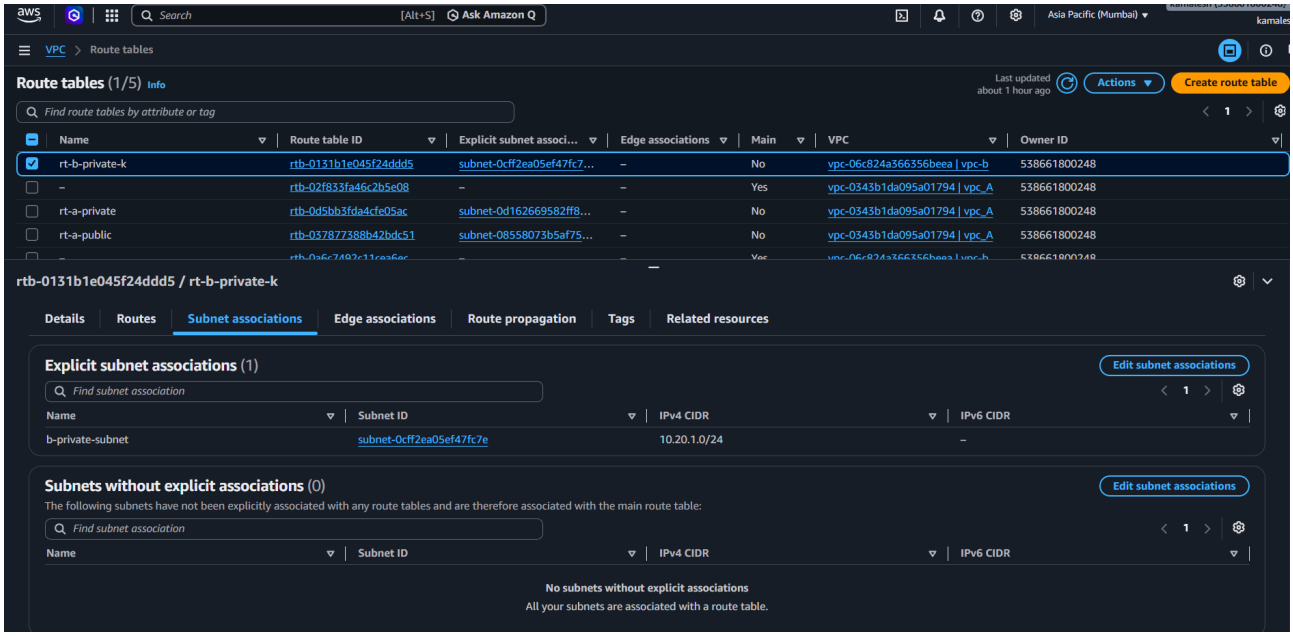
Evidence 7 — App-A Private Subnet Association

Shows rt-a-private explicitly associated with subnet a-private, CIDR 10.10.2.0/24.



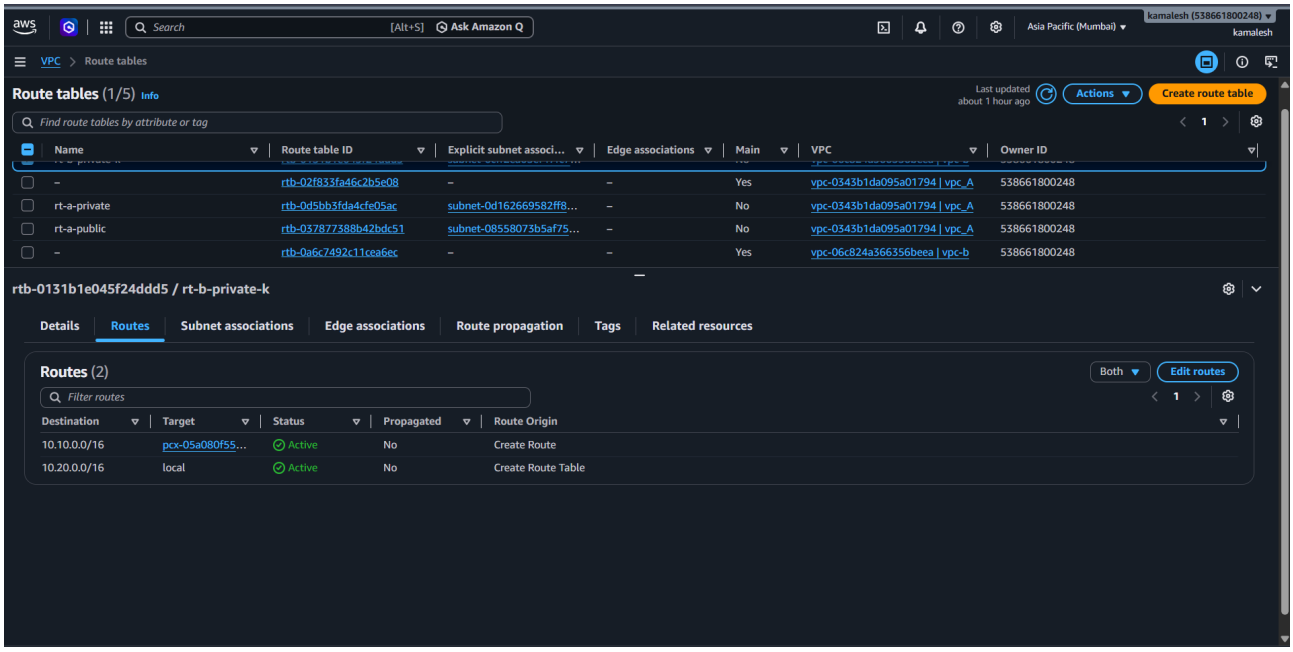
Evidence 8 — App-B Private Subnet Association

Shows rt-b-private-k explicitly associated with b-private-subnet, CIDR 10.20.1.0/24.



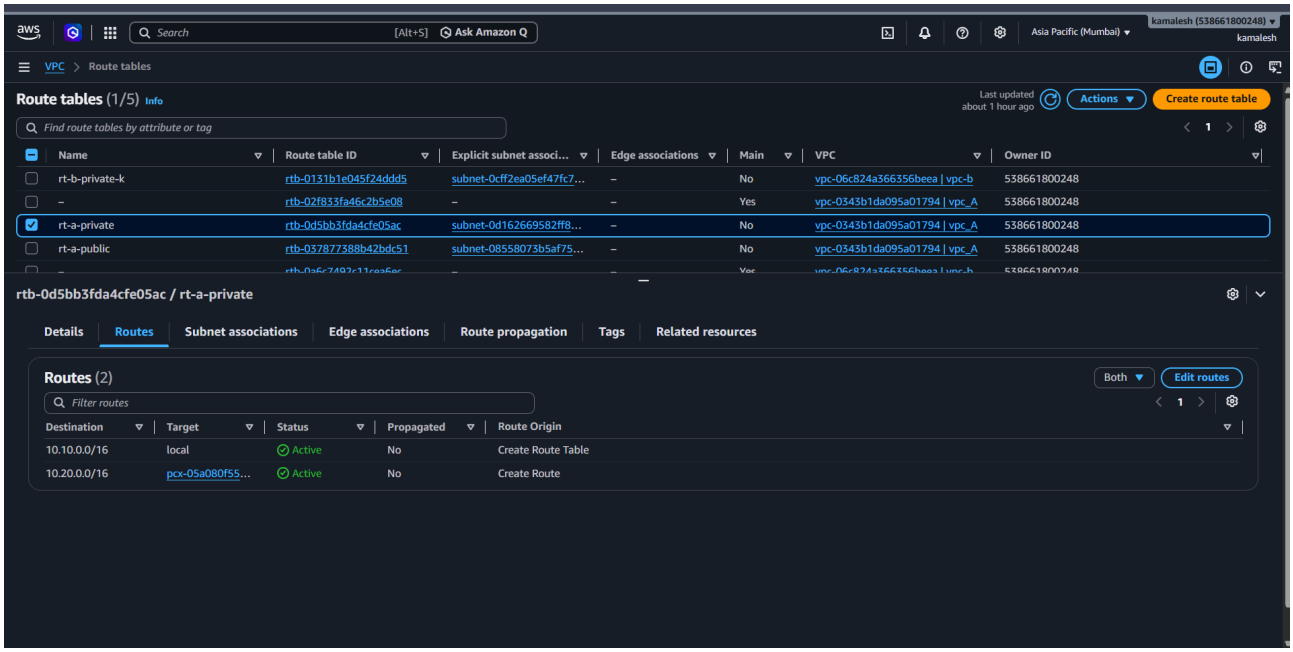
Evidence 9 — App-B Private Route Table

Shows route 10.10.0.0/16 to peering connection and local 10.20.0.0/16 route; both Active.



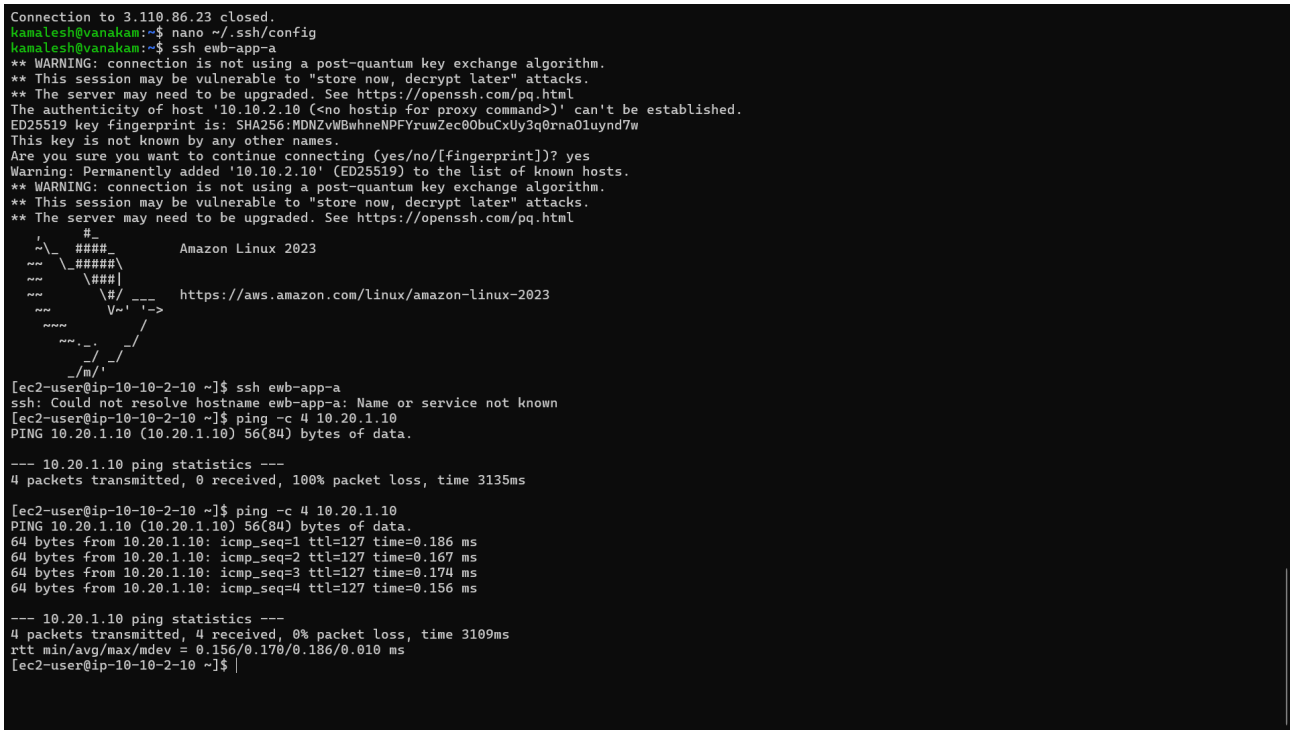
Evidence 10 — App-A Private Route Table

Shows local 10.10.0.0/16 route and 10.20.0.0/16 to peering connection; both Active.



Evidence 11 — Successful App-A to App-B Ping

Terminal evidence shows ping to 10.20.1.10 with 4 packets received and 0% packet loss.



7. Learning Summary

This project helped demonstrate VPC peering, CIDR planning, route-table configuration, subnet associations, security-group access control and private-IP connectivity testing. A peering connection alone is not sufficient: both VPCs need routes to the remote CIDR, and security groups must allow the required traffic.

8. Final Notes

This report presents the screenshots and successful test output currently available. Keep the original screenshots as supporting evidence. Before deleting lab resources, ensure the required project evidence has been saved and check for any remaining billable resources.